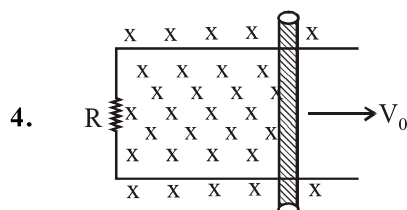


ELECTROMAGNETIC INDUCTION

1. Eddy currents are generated in :-
 (1) insulator
 (2) conductor
 (3) both
 (4) none of these
2. When a high frequency current is passed through a coil which surrounds the metal the _____ generated in metal produces _____ temperature sufficient to _____ it :-
 (1) induced emf, low, freeze
 (2) induced current, low, melt
 (3) eddy current, high, melt
 (4) mechanical energy, high, freeze

3. If a copper plate is allowed to oscillate between (N) pole & (S) pole of a magnet like a simple pendulum, then :-
 (1) it will oscillates for ∞ time
 (2) it's motion is accelerated
 (3) it will not oscillate
 (4) it's oscillation is damped



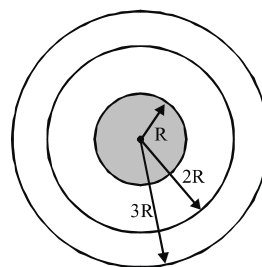
If this rod length of rod $\perp$ to magnetic field is released with initial velocity V_0 then :-

- (1) It will continue to move with constant velocity
 - (2) It will accelerate
 - (3) It will stop after sufficiently long time
 - (4) None of these
5. Total amount of work done in establishing the current I in coil of inductance L is :-

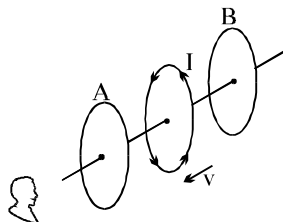
$$(1) W = \int_0^I I(dI) \quad (2) W = \int_0^I LI^2(dI)$$

$$(3) W = -\int_0^I I(dI) \quad (4) W = \int_0^I LI(dI)$$

6. The dimension of the ratio of magnetic flux and the resistance is equal to that of :
 (1) induced emf (2) charge
 (3) inductance (4) current
7. A uniform magnetic field $\vec{B}$ exists in circular region of radius R as shown which is decreasing with time at a constant rate. There is a concentric wooden ring which have uniformly distributed charge Q and radius $2R$ and another concentric uncharged copper ring of radius $3R$, then select **INCORRECT** alternative : (Assume rings are placed on frictionless surface)

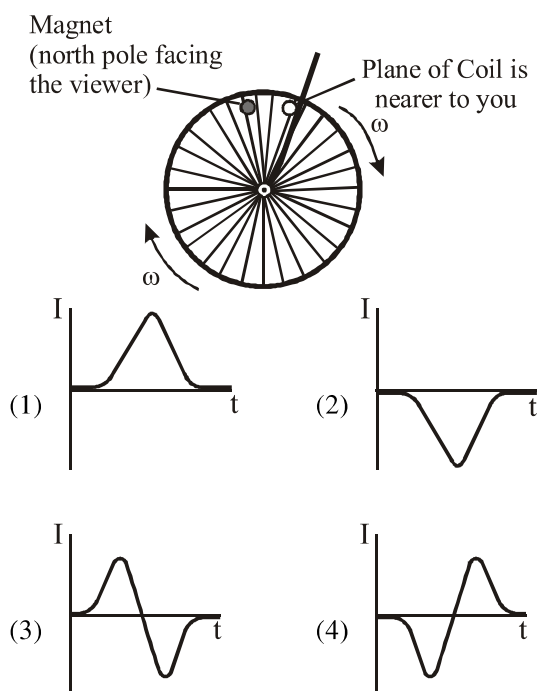


- (1) There is no induced current in wooden ring.
 - (2) Induced electric field in wooden ring is more in magnitude than copper ring.
 - (3) Induced electric field in copper ring is more in magnitude than wooden ring.
 - (4) Wooden ring will start rotating.
8. Three wire loops and an observer are positioned as shown in the figure. From the observer's point of view, a current I flows counterclockwise in the middle loop, which is moving towards the observer with a velocity v . Loops A and B are stationary. This same observer would notice that

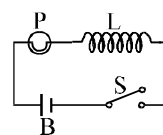


- (1) clockwise currents are induced in loops A and B
- (2) counterclockwise currents are induced in loops A and B
- (3) a clockwise current is induced in loop A, but a counterclockwise current is induced in loop B
- (4) a counterclockwise current is induced in loop A, but a clockwise current is induced in loop B

9. In a bicycle speedometer, a bar magnet is attached to the spokes of the wheel and a coil is attached to the frame so that the north pole of the magnet moves past it once for every revolution of the wheel. As the magnet moves past the coil, a pulse of current is induced in the coil. A computer then measures the time between pulses and computes the bicycle's speed. Figure shows the magnet about to move past the coil. Which of the graphs shows the resulting current pulse? Take counterclockwise in figure current to be positive.



10. A bound wire in the form of a coil on an iron core behaves as an inductor. A rheostat having similar structure has :
- only resistance but no inductance.
 - both resistance and inductance.
 - it offers resistance only to DC and impedance $Z > R$ to AC.
 - it doesn't have iron core so no inductance.
11. In figure, a lamp P is in series with an iron-core inductor L. When the switch S is closed, the brightness of the lamp rises relatively slowly to its full brightness than it would do without the inductor. This is due to



- the low resistance of P
- the induced-emf in L
- the low resistance of L
- the high voltage of the battery B